

Docker Cheat Sheet

Beginner-friendly quick revision — only the most useful concepts and commands

1. Docker Basics

Concept	Simple meaning
Docker	Platform for building and running applications in containers.
Container	A running/created instance of an image.
Image	Package/template used to create containers.
Dockerfile	Instructions used to build an image.
Registry	Stores and distributes images.
Volume	Persistent storage for container data.
Network	Allows containers to communicate.

2. Docker Workflow

Step	What happens
1	Write application code + Dockerfile.
2	Build an image.
3	Run a container from the image.
4	Check status and logs.
5	Push image to a registry when sharing/deploying.

3. Dockerfile — Easy Must-Know Instructions

Instruction	Use
FROM	Choose the base image.
WORKDIR	Set the working directory.
COPY	Copy application files into the image.
RUN	Run a command during image build.
ENV	Set an environment variable.
EXPOSE	Document the application port.
CMD	Default command when the container starts.

4. Most Important Commands

Command	What it does
<code>docker --version</code>	Show Docker version.
<code>docker images</code>	List images.
<code>docker ps</code>	List running containers.
<code>docker ps -a</code>	List all containers.
<code>docker pull IMAGE</code>	Download an image.
<code>docker build -t APP:TAG .</code>	Build an image.
<code>docker run IMAGE</code>	Create and start a container.
<code>docker run -d -p 8080:80 IMAGE</code>	Run in background + map port 8080 to 80.
<code>docker stop CONTAINER</code>	Stop a container.
<code>docker start CONTAINER</code>	Start a stopped container.
<code>docker rm CONTAINER</code>	Remove a stopped container.

5. Logs, Shell & Troubleshooting

Command	What it does
<code>docker logs CONTAINER</code>	View container logs.
<code>docker logs -f CONTAINER</code>	Follow logs live.
<code>docker exec -it CONTAINER sh</code>	Open a shell inside a running container.
<code>docker inspect CONTAINER</code>	Show detailed container information.
<code>docker stats</code>	Show live resource usage.

6. Ports, Environment & Storage

Feature	Simple meaning / example
Port mapping	-p HOST:CONTAINER — e.g. 8080:80.
Environment	-e KEY=VALUE — pass configuration to a container.
Named volume	-v NAME:/path — keep data outside the container lifecycle.
Bind mount	-v HOST:/path — mount a host directory.

7. Images & Registries

Command	What it does
<code>docker tag IMAGE REPO:TAG</code>	Give an image a repository tag.
<code>docker login</code>	Log in to a registry.
<code>docker push REPO:TAG</code>	Upload an image.
<code>docker pull REPO:TAG</code>	Download an image.

8. Docker Compose — Beginner Essentials

Command / File	Simple meaning
<code>compose.yml</code>	Defines services, networks and volumes.
<code>docker compose up</code>	Create/start the services.
<code>docker compose up -d</code>	Start services in background.
<code>docker compose down</code>	Stop and remove Compose resources.
<code>docker compose ps</code>	Show Compose containers.
<code>docker compose logs</code>	Show service logs.

9. Basic Cleanup

Command	Meaning
<code>docker container prune</code>	Remove stopped containers.
<code>docker image prune</code>	Remove unused/dangling images.
<code>docker volume prune</code>	Remove unused volumes.

10. Docker + Kubernetes

Docker	Builds, packages and runs application containers.
Kubernetes	Orchestrates containers across a cluster.
Image	Package pulled by a container runtime.
Registry	Stores and distributes images.

11. Security — Remember These

Practice	Why
Use trusted/minimal images	Reduces unnecessary software and attack surface.
Scan images	Find known vulnerabilities.
Never hard-code secrets	Avoid credentials inside images.
Prefer non-root containers	Reduces impact of a compromise.

Quick revision: Dockerfile → Build → Image → Registry → Run → Container → Logs → Stop/Remove